Research Report – Quantum Computing and Cryptography: A Security Disruption

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***Executive Summary*— The coming of quantum computing is expected to break many security systems that rely on current cryptography. This includes classical cryptography algorithms like RSA and ECC that secure current global communications, financial systems, and digital infrastructures. These security systems would become vulnerable to quantum attacks, especially from algorithms like Shor's and Grover's. Although quantum computers aren't widely accessible currently, cybercriminals may already be harvesting and collecting encrypted data for future decryption. To defend against this risk, people around the world and the cryptography community are developing post-quantum cryptography (PQC), which are algorithms resistant to both classical and quantum attacks. In this report explored the foundation of PQC, NIST's standardisation, and current strategies for organisations to migrate to quantum-safe protocols. As well as outlined practical challenges and current issues. Organisations are encouraged to prepare and act now to avoid those threats in the future once quantum computing becomes more accessible. Big companies such as NIST and IBM are working closely together to create a standard for PQC, ensuring that everyone is safe from threats of quantum attacks.**

Keywords*— *Post Quantum Cryptography (PQC), Quantum Computing, Cryptographic Algorithms, Shor's Algorithm, Grover's Algorithm, NIST Standardization, Hybrid Cryptography, Quantum Safe Protocols, Side-Channel Attacks, Cryptographic Migration.

I. INTRODUCTION

In today's digital age, cryptography forms the backbone of secure communication. Whether it's protecting online banking transactions, securing corporate data, or authenticating difital identities. Cryptographic algorithms are deeply embedded in the foundation of internet infrastructure where trusts in online systems relies on. The security of widely used cryptography algorithms like RSA [1] because classical computer can't efficiently solve its math problems. On another hand, symmetric-key algorithms like AES are designed to resist brute-force attacks [2].

However, the coming of quantum computing creates fundamentally challenge to the security with these cyptography algorithms. Unlike classical computing that process information or data in binary bits (0 and 1), quantum computers use quantum bits or qubits, which can exist in multiple states simultaneously due to the principle of superposition [3]. Furthermore, quantum entanglement enables qubits to interact with each other in powerful ways, which allows quantum computers to perform calculations a lot faster than classical computers.

Therefore, this leap in computational power introduces new vulnerabilities, where Shor's Algorithms can efficiently solve difficult integer factorization problems [4], making RSA effectively obsolete in a post-quantum world. Similarly, Grover's Algorithms provides a quadratic speed up for searching unsorted databases which brute-force attacks against symmetric algorithms like AES by reducing their effective security by half [5]. These breakthroughs threaten the very foundation of cryptographic security used today.

Even though large-scale quantum computers are not yet available, attackers may already be preparing for the future. Attackers may realize that in the future when quantum computers become more accessible, they could use it to break the encryption. They may be collecting encrypted data now for the future uses, this called "Harvest Now, Decrypt Later" strategies [6]. Therefore, organizations start looking for solutions to defences against these potential attacks in the future. This is where Post Quantum Cryptography (PQC) comes in, an algorithm designed to remain secure against both classical and quantum computers [7].

This report explores the foundations of quantum-resistant cryptography, the algorithms and frameworks being developed to secure systems in a post-quantum era, and the technical and practical challenges that remain. By understanding the transition process, we can better prepare for the shift toward quantum-resilient security in critical applications.

II. THE THREAT OF QUANTUM COMPUTING

A new generation of computing is possible with the laws of quantum mechanics, which represent a new paradigm in computation like quantum computing. It unlocked and upgraded huge computational power compared to classical computers that use bits (which are either 0 or 1). Instead, quantum computers use qubits, or quantum bits. One of four key principles of quantum mechanics, superposition, enables qubits to be in both states (0 or 1) at the same time. As well as entanglement, qubits can influence each other instantaneously, allowing quantum systems to perform certain calculations much more efficiently than classic computers. This results in many possibilities quantum algorithms can explore, which may lead to solutions to problems previously thought impossible to solve.

A. Quantum Threat: Shor's and Grover's Algorithms

In the quantum computing world, there are two key algorithms that create a great threat to the current reliable cryptographic security of classic computing: Shor's and Grover's. Shor's algorithm was introduced in 1994; it could solve difficult integer factorization problems, which makes RSA ineffective. Grover's algorithm is able to solve brute-force search problems at a quadratic speed; it makes symmetric algorithms like AES less effective by half [5]. For instance, using AES-128 would equal 64 bits of quantum security in terms of effectiveness, making it vulnerable to quantum-accelerated attacks. These quantum capabilities threaten the confidentiality, authenticity, and integrity of digital communication if new quantum-safe systems are not adopted.

B. *Real World Impact on Protocols*

The rise in quantum computing places a wide range of systems at risk. Internet protocols such as TLS/SSL, IPSec VPNs, and SSH, which depend on RSA or ECC for key exchange and authentication, would be rendered insecure [8] [9]. This not only affects web browsing and email encryption but also financial transactions, secure communications, digital contracts, and even blockchain technologies like Bitcoin. Digital signatures, which ensure trust and identity in online systems, would also be compromised. As a result, security needs to be re-engineered to be safe and secure after quantum computing becomes more accessible.

C. *Urgency: Harvest Now, Decrypt Later (HNDL)*

Even though the accessibility to quantum computing is relatively small today, hackers or cybercriminals may be collecting current encrypted data for future uses after quantum computers become more accessible. In a "Harvest Now, Decrypt Later" strategy, cybercriminals may use encrypted data collected from the current date and decrypt that data in the future [6]. Much sensitive data with long-term values, such as military intelligence, healthcare records, or financial archives, could be in critical threat. Recognising this, agencies such as the National Institute of Standards and Technology (NIST) have issued warnings and are actively developing strategies for transitioning to post-quantum cryptography before quantum decryption becomes feasible [10].

III. Introduction to Post Quantum Cryptography

Post-Quantum Cryptography (PQC) is a new generation of cryptographic algorithms that were designed to resist attacks from quantum computers. PQC algorithms don't rely on problems like integer factorization or discrete logarithm problems for security, such as RSA and ECC, because Shor's algorithm could solve them. Instead, PQC design was based on mathematical problems that are believed to be difficult for both classical and quantum computers to solve. Which keeps current sensitive data safe for after quantum computers become more accessible. Importantly, PQC does not require quantum hardware and is implemented using classical computing, making it deployable in current systems as a proactive defence [11].

Therefore, the world urgently needs some kind of quantum-resistant cryptography to prevent the potential collapse of widely used encryption protocols once quantum computing becomes available. Protocols like RSA and ECC, which protect everything from the internet connection to digital identities, can be broken by quantum algorithms. This means systems dependent on them would become insecure virtually overnight, because many sensitive data assets need to remain for years and decades. As a result, those future risks could be considered as present today. Transitioning to PQC before quantum decryption becomes a reality is essential to ensure long-term data confidentiality, integrity, and trust across global digital infrastructure.

A. *Mathematical Foundation of PQC*

Post-quantum cryptography relies on alternative mathematical foundations believed to be resistant to both classical and quantum attacks. Key categories include:

1) *Lattice Based Cryptography*

Lattice-based cryptography is built on complex geometric structures called lattices, which are grids of points in multi-dimensional space. Security comes from problems like Learning With Errors (LWE), Shortest Vector Problem (SVP), and Closest Vector Problem (CVP) [14]. These problems remain hard even for quantum computers. Lattice-based schemes are fast and efficient, and they support both encryption and digital signatures. Two of the leading candidates standardised by NIST are ML-KEM (CRYSTALS-KYBER) and ML-DSA (CRYSTALS-DILITHIUM) [13].

2) *Hash Based Cryptography*

Hash-based cryptography uses only cryptographic hash functions. The idea is to build secure digital signatures by chaining hashes in a way that is one-way and hard to reverse or find two inputs with the same hash. Hash functions are widely trusted and offer very strong security. However, it is typically only useful for digital signatures, not encryption. The leading candidate standardised by NIST is SLH-DSA (SPHINCS+) [13].

3) *Code Based Cryptography*

Code-based cryptography is based on the problem of decoding error-correcting codes [15]. The basic idea is to disguise a structured code in a random looking way and then use the hidden structure to decode it efficiently, while attackers cannot. These systems offer very strong theoretical security, even against quantum attacks. This method is best suited for applications where key size is not a major constraint.

B. *NIST PQC Standardization*

To guide global adoption, the National Institute of Standards and Technology (NIST) initiated a competition-like standardisation project in 2016 to evaluate and select PQC algorithms. NIST selected four algorithms for further evaluation from 69 submissions, which are CRYSTALS-KYBER, CRYSTALS-DILTHIUM, Falcon, and SPHINCS+ in 2022 [16]. In 2024, the collaboration of two industry partners, IBM and NIST, published the world's first three PQC standards. IBM researchers developed ML-KEM, ML-DSA, and SLH-DSA post-quantum cryptographic algorithms in collaboration with several industry partners [16].

IV. Quantum-Safe Protocols and Frameworks

There are many new cryptographic protocols and frameworks that are being developed to protect current systems from future quantum threats. These are designed to replace or strengthen the parts of communication systems that rely on RSA, ECC, or other quantum-vulnerable algorithms. So, using a quantum-safe algorithm from the NIST PQC competition is a strong approach. For instance, CRYSTALS-KYBER is designed for encryption and key exchange, while CRYSTALS-DILITHIUM, Falcon, and SPHINCS+ are used for digital signatures. These new algorithms are being prepared for wide adoption and will eventually replace older algorithms in systems like web browsers, secure email, and cloud services.

Even though the world is still in the early stages of adoption, a common method right now is to use hybrid cryptography [17]. Which means combining classical and quantum-safe algorithms in the same protocol, so if one fails, the other still protects the data. Google tested this in its Chrome browser with an experiment called CECPQ1, and later CECPQ2, using both X25519 (ECC) and post-quantum key exchange (KYBER) [18] [19].

There are also open-source projects helping developers and companies test and adopt PQC. Open Quantum Safe (OQS) is an open-source project that aims to support the transition to quantum-resistant cryptography. OQS provides two main tools, such as liboqs and open-source C library for quantum resistance [20]. These protocols and frameworks are a big part of the preparation for post-quantum threats.

V. Current Challenges and Open Issues

A. *Performance Impact*

Post-quantum algorithms often trade classical speed for quantum resistance. It requires more computational resources, increasing key and ciphertext sizes, than classical cryptography. Usually, the system will run slower and need more space to store a larger key and ciphertexts, which taxes bandwidth and memory-constrained hardware. Organisations may need to carefully evaluate whether their current system hardware could support the increased computational load of PQC or not, as many may require hardware upgrades. As a result, increase costs and pose a significant deployment challenge [21].

B. *Migration and Interoperability with Legacy Systems*

There is huge potential that not all network elements and partner systems will upgrade to PQC at the same time. The PQC-based system tries to communicate with the legacy system; a secure connection couldn’t be established. A single non-upgraded component can block migration and create deployment bottlenecks. A hybrid cryptographic modes solution for migration could be a good choice, but this adds complexity and potentially slows down the transition. Successful migration may require systems to be upgraded simultaneously or use hybrid modes for backwards compatibility [21].

C. *Security Risks*

The migration from legacy systems to PQC-based systems raises security concerns because new algorithms have not been tested in real-world deployments, unlike RSA and ECC, which have been around for decades. This increases the risks of undiscovered weaknesses and implementation flaws. Also, cryptanalysis and side-channel attacks have already been found during standardisation. Ensuring side-channel resistance is essential; cryptographic operations must not leak secrets through timing, power, or memory access. PQC also introduces new challenges in key management and failure modes. Early deployments have revealed issues like firewalls or middleware breaking due to large key sizes, and misconfigurations in hybrid modes or certificate handling can create new vulnerabilities. [21].

VI. Future Directions and Migration Strategies

A. *Hardware and Performance Optimisation*

To reduce the performance burden of PQC, researchers and vendors are working on optimising algorithms and implementations. Which includes tuning parameters to balance security and speed, developing more efficient software libraries, and using hardware acceleration. Choosing algorithms with smaller keys and faster operations, like KYBER and DILITHIUM, can also help systems with limited resources.

B. *Smooth Migration and Hybrid Modes*

Organisations should adopt hybrid cryptography to ensure a smooth transition from legacy systems to PQC-based systems, which combines classical and post-quantum algorithms. This allows backwards compatibility while gradually increasing security. However, the migration process should start deploying in non-customer-facing areas first, then expanding as standards and interoperability improve over time. Organisations should also follow structured migration plans that align with regulations and upgrade cycles, especially when PQC becomes part of 5G-Advanced and future 6G networks. Standards from NIST and others will shape these transitions from 2024 onward, making early planning critical. [21].

C. *Secure Implementation and Auditability*

Security remains a key concern as many post quantum algorithms are still new and haven't been tested in large-scale, real-world environments. Developers must follow secure coding practices, especially their implementation are resistant to side-channel attacks to reduce the risk of vulnerabilities. Using constant-time operations, masked computations, and code that has been formally verified helps protect private keys from leaking during execution. In the future, more tools and standards will be developed to improve the quality and auditability of PQC base systems.

D. *Policy, Training, and Future Readiness*

To stay ahead of the quantum threat, organisations need to plan early. Conducting a detailed list or map of where and how cryptography is used is very important for system maintainability and identifying high-priority assets that may be at risk. Importantly, many PQC upgrades can be deployed through software updates, meaning expensive hardware replacements are not always necessary [21].

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